

D568

Health Equity and Social Determinants of Health

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
-----------------------------	---	------------------------	----------------------------------

Items

1.

A public health nurse in a county that recently opened a new library notes that one neighborhood has no sidewalks, few safe parks, and high levels of air pollution from a nearby highway. Residents there have higher asthma and obesity rates. The county fair is held every August. Which SDOH domain do these conditions best fit?

- A. Neighborhood and built environment.
- B. Health care access and quality.
- C. Education access and quality, because of the new library.
- D. Individual lifestyle choices, not a social determinant.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A school district that just repaved its bus lot gives every student the same \$50 voucher for school supplies. Some families have no reliable internet at home, and some students need glasses they cannot afford. The superintendent is a former science teacher. Which change would move the program from equality toward equity?

- A. Raise the voucher to \$75 for every student.
- B. Provide additional support such as internet hotspots and eye exams to students who face those barriers.
- C. Keep the equal voucher, since identical treatment is always fair.
- D. Give the largest vouchers to families that already have internet and glasses.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A clinic gives every patient the same printed English-language instructions after a visit. What does this approach represent, and what is its main limitation?

- A. An upstream intervention.
- B. Equality; patients with limited English or literacy may not benefit.
- C. Equity; everyone gets the same information.
- D. Justice; no limitation exists.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A state health report with a new online dashboard shows two differences. Adults over 75 have higher rates of hip fracture than adults aged 30–40. Residents of low-income neighborhoods have higher rates of preventable hospitalization because nearby clinics closed and transit is limited. The report was released in the spring. Which difference is best described as a health inequity?

- A. Both, because every group difference is an inequity.
- B. Neither, because both are just differences in rates.
- C. The higher preventable hospitalization rate in low-income neighborhoods.
- D. The higher hip fracture rate in adults over 75.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

What would have to be true for a difference in diabetes rates between two groups to be called a health inequity rather than only a disparity?

- A. The difference stems from avoidable, unjust conditions such as unequal access to healthy food or care.
- B. The difference is measurable.
- C. The difference is caused by genetic factors alone.
- D. The difference is larger than 10 percentage points.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A city with a well-known minor-league baseball team has high rates of childhood lead poisoning in older rental housing. The health department has a new logo. Which intervention is most upstream?

- A. Hand out brochures on lead dangers at clinics.
- B. Treat children with high lead levels using chelation.
- C. Pass and enforce a rental code requiring lead-safe housing before units can be rented.
- D. Screen children's blood lead levels at checkups.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

Which of these is a downstream intervention?

- A.** Zoning rules that keep diesel truck routes away from schools.
- B.** Raising the minimum wage.
- C.** Any program run by a public agency.
- D.** A hospital program that manages uncontrolled asthma in children after emergency visits.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A regional health coalition that meets in a converted train station has funding for one strategy to reduce tooth decay across 200,000 residents. Its chair is a retired dentist. According to the health impact pyramid, which option will likely have the broadest population impact?

- A.** Offer one-on-one brushing counseling at dental visits.
- B.** Run a public service campaign on sugary drinks.
- C.** Hire more dentists, since clinical care always has the largest effect.
- D.** Fluoridate the community water supply.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

County A, which has a large state university, had 45 opioid overdose deaths last year in a population of 30,000. County B, known for its apple orchards, had 40 deaths in a population of 80,000. What are the death rates per 100,000 and the rate ratio of A to B?

- A.** A and B are about equal, since 45 and 40 deaths are similar.
- B.** A = 45, B = 40; rate ratio 1.1.
- C.** A = 150, B = 50; rate ratio 0.33.
- D.** A = 150, B = 50; rate ratio 3.0.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A town of 25,000 has a tuberculosis rate of 120 per 100,000. How many cases does that represent?

- A.** 30
- B.** 480
- C.** 120
- D.** 3,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A health researcher at a university with a famous marching band maps asthma rates in a city and finds that the highest rates today fall almost exactly within neighborhoods that were rated “hazardous” on 1930s federal lending maps. Those areas have more highways, fewer trees, and older housing. The city recently hosted a film festival. What best explains the pattern?

- A.** Coincidence, since policies from the 1930s cannot affect health today.
- B.** Interpersonal bias by today’s local doctors.
- C.** Structural racism through historical redlining that shaped investment and environmental exposure.
- D.** Individual residents’ lifestyle choices.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A university team with a new research building wants to reduce food insecurity in a rural town. The team’s lead has published widely. Which approach best reflects community-based participatory research?

- A.** Let the lead researcher decide everything, since expertise guarantees relevance.
- B.** Form a partnership in which residents help define the question, design the study, interpret findings, and decide how results are used.
- C.** Survey residents about food access and publish the results.
- D.** Design the intervention at the university and invite residents to a launch event.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A public health nurse in a county ~~that recently opened a new library~~ notes that one neighborhood has no sidewalks, few safe parks, and high levels of air pollution from a nearby highway. Residents there have higher asthma and obesity rates. ~~The county fair is held every August.~~ Which SDOH domain do these conditions best fit?

Target word: SDOH domain **Decides:** Physical features of where people live.

A	Correct	Sidewalks, parks, and air quality are features of the physical environment.
B	Adjacent category	Nothing described is about access to clinical care.
C	Camouflage	The library is irrelevant to the conditions described.
D	Inverted operation	Frames environmental conditions as personal choice.

2. Answer B Concept 2

A school district ~~that just repaved its bus lot~~ gives every student the same \$50 voucher for school supplies. Some families have no reliable internet at home, and some students need glasses they cannot afford. ~~The superintendent is a former science teacher.~~ Which change would move the program from equality toward equity?

Target word: equity adjustment **Decides:** Resources matched to differing needs.

A	Dropped element	Still identical for all; ignores differing needs.
B	Correct	Equity allocates by need so all students have a fair chance.
C	Absolute language	Equal inputs can leave unequal barriers in place.
D	Inverted operation	Sends more to those with fewer barriers.

3. Answer B Concept 2 · transfer

A clinic gives every patient the same printed English-language instructions after a visit. What does this approach represent, and what is its main limitation?

Target word: classify a same-for-all approach **Decides:** Identical treatment regardless of need.

A	Adjacent category	Discharge instructions are downstream care.
B	Correct	Same resource for all without adjusting for barriers.
C	Confusion set	That describes equality.
D	Absolute language	Ignores the language barrier.

4. Answer C Concept 3

A state health report ~~with a new online dashboard~~ shows two differences. Adults over 75 have higher rates of hip fracture than adults aged 30–40. Residents of low-income neighborhoods have higher rates of preventable hospitalization because nearby clinics closed and transit is limited. ~~The report was released in the spring.~~ Which difference is best described as a health inequity?

Target word: disparity vs. inequity **Decides:** Avoidable and unjust vs. largely biological.

A	Absolute language	Only avoidable, unjust differences are inequities.
B	Dropped element	Drops the cause, which is what separates inequity from disparity.
C	Correct	Driven by avoidable, unfair conditions such as clinic closures and poor transit.
D	Confusion set	A disparity driven mainly by age-related biology, not by unjust conditions.

5. Answer A Concept 3 • transfer

What would have to be true for a difference in diabetes rates between two groups to be called a health inequity rather than only a disparity?

Target word: criterion for inequity **Decides:** Avoidable and unjust causes.

A	Correct	Inequity requires avoidable and unfair causes.
B	Adjacent category	That only makes it a disparity.
C	Inverted operation	Purely biological causes point away from inequity.
D	Wrong stem number	Size alone does not make a difference unjust.

6. Answer C Concept 4

A city ~~with a well-known minor league baseball team~~ has high rates of childhood lead poisoning in older rental housing. ~~The health department has a new logo.~~ Which intervention is most upstream?

Target word: most upstream option **Decides:** Changes the conditions that cause exposure.

A	Confusion set	Education asks individuals to act but leaves the housing unchanged.
B	Inverted operation	Treatment is the furthest downstream step.
C	Correct	Removes the cause for everyone in affected housing.
D	Adjacent category	Screening finds exposure after it occurs—downstream.

7. Answer D Concept 4 • transfer

Which of these is a downstream intervention?

Target word: identify downstream — reverse direction **Decides:** Treating people after illness appears.

A	Confusion set	Changes environmental exposure—upstream.
B	Inverted operation	Changes economic conditions—upstream.
C	Absolute language	Who runs a program does not decide its position upstream or downstream.
D	Correct	Care delivered after illness has occurred.

8. Answer D Concept 5

A regional health coalition ~~that meets in a converted train station~~ has funding for one strategy to reduce tooth decay across 200,000 residents. ~~Its chair is a retired dentist.~~ According to the health impact pyramid, which option will likely have the broadest population impact?

Target word: highest-reach option **Decides:** Changing the context so the healthy option is the default.

A	Inverted operation	Counseling is at the top of the pyramid—least population reach.
B	Adjacent category	Education still depends on individual behavior change.
C	Absolute language	Clinical care reaches individuals, not whole populations.
D	Correct	Changes the context for everyone with no individual effort required.

9. Answer D Concept 6

County A, ~~which has a large state university~~, had 45 opioid overdose deaths last year in a population of 30,000. County B, ~~known for its apple orchards~~, had 40 deaths in a population of 80,000. What are the death rates per 100,000 and the rate ratio of A to B?

Target word: rates and rate ratio **Decides:** Rate = deaths ÷ population × 100,000.

A	Dropped element	Compares counts and drops population size.
B	Wrong stem number	Uses the raw counts as rates.
C	Inverted operation	Divides B by A instead of A by B.
D	Correct	$45/30,000 \times 100,000 = 150$; $40/80,000 \times 100,000 = 50$; $150/50 = 3.0$.

10. Answer A Concept 6 · transfer

A town of 25,000 has a tuberculosis rate of 120 per 100,000. How many cases does that represent?

Target word: solve for count from a rate **Decides:** Cases = rate × population ÷ 100,000.

A	Correct	$120 \times 25,000 / 100,000 = 30$.
B	Inverted operation	Divides 120 by 0.25 instead of multiplying by 0.25.
C	Wrong stem number	Uses the rate as the count.
D	Dropped element	Multiplies 120 by 25 (the population in thousands) and drops the final ÷ 100 that finishes the conversion to the 100,000 base.

11. Answer C Concept 7

A health researcher at a university ~~with a famous marching band~~ maps asthma rates in a city and finds that the highest rates today fall almost exactly within neighborhoods that were rated “hazardous” on 1930s federal lending maps. Those areas have more highways, fewer trees, and older housing. ~~The city recently hosted a film festival.~~ What best explains the pattern?

Target word: type of determinant **Decides:** A past policy shaped investment and environment for generations.

A	Absolute language	Structural effects persist across generations.
B	Adjacent category	Interpersonal bias may exist, but the map overlap points to structural causes.
C	Correct	An institutional policy produced lasting neighborhood conditions tied to health.
D	Inverted operation	Ignores the policy-driven environment.

12. Answer B Concept 8

A university team ~~with a new research building~~ wants to reduce food insecurity in a rural town. ~~The team's lead has published widely.~~ Which approach best reflects community-based participatory research?

Target word: true partnership **Decides:** Community shares power across every stage.

A	Absolute language	Expertise does not replace community knowledge of local needs.
B	Correct	Equal partnership across all phases defines CBPR.
C	Confusion set	Residents serve only as data sources.
D	Dropped element	Drops community input in design and decision-making.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	7	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	The five SDOH domains	Social determinants are commonly grouped into economic stability, education access and quality, health care access and quality, neighborhood and built environment, and social and community context.	Treating SDOH as only individual choices or only medical care.	A condition of where people are born, live, learn, work, or play.
2	Equality vs. equity	Equality gives everyone the same resources; equity allocates resources according to need so that everyone has a fair chance to reach good health.	Assuming identical treatment produces fair outcomes.	Same amount for all vs. more for those with greater barriers.
3	Disparity vs. inequity	A health disparity is any measurable difference between groups; a health inequity is a difference that is avoidable, unnecessary, and unjust.	Calling every group difference an inequity, including those driven by biology such as age.	Is the difference rooted in unfair, changeable conditions?
4	Upstream vs. downstream interventions	Upstream interventions change the social, economic, and environmental conditions that cause illness; downstream interventions treat people after problems appear.	Labeling a better clinic service as upstream.	Policy, housing, income, or environment vs. treatment of individuals.
5	The health impact pyramid	Interventions at the base (socioeconomic factors) and changing the context so healthy choices are the default reach the most people with the least individual effort; counseling and education at the top reach fewer and require the most individual effort.	Assuming health education is the highest-impact strategy.	Comparing a policy or environmental change with an education campaign.
6	Rates, not counts, compare burden	Compare groups using rates (cases ÷ population × a standard base such as 100,000), then a rate ratio (rate in one group ÷ rate in the reference group).	Concluding the group with more cases has the greater burden.	Case counts from populations of very different sizes.
7	Structural determinants	Structural racism and other structural determinants work through laws, policies, and institutional practices (such as historical redlining) that shape where resources go across generations, even without individual prejudice.	Attributing group differences only to individual behavior or individual bias.	Present-day disparities that map onto past policy boundaries.

#	Concept	Rule	Trap	Tell
8	Community engagement and CBPR	Community-based participatory research treats community members as equal partners in defining problems, designing, carrying out, and using the results, not just as data sources.	Calling a survey of residents “community partnership.”	Who sets the question and who shares decisions and results.

Course scope (public description): Health Equity and Social Determinants of Health examines the social determinants of health (SDOH) as underlying factors that contribute to health inequity in populations and communities and their effect on health outcomes. This course will help students understand the evidence-based strategies and approaches that promote health equity.